ApleShahar: Cloud-Native Civic Grievance & Municipal Infrastructure Platform

A Mini-Project Report Submitted For
Cloud Computing Lab In
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In
COMPUTER ENGINEERING
(Semester V)

By

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CERTIFICATE

This is to certify that the mini-project entitled "ApleShahar: Cloud-Native Civic Grievance & Municipal Infrastructure Platform" is a bonafide work of Arnav Patade (10910), Arlen DMello (10909), and Hansy Rodrigues (10633) submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Engineering in Computer Engineering (Semester V) under the course Cloud Computing Lab.

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DECLARATION

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ABSTRACT

Urban infrastructure management in expanding metropolitan regions such as the Vasai-Virar City Municipal Corporation (VVCMC) and Mumbai Metropolitan Region (MMR) faces critical systemic challenges: unmonitored road hazards, illegal waste dumping, damaged public utilities, and an absence of transparent tracking mechanisms for civic complaints. Traditional municipal grievance systems rely on paper-based records or legacy monolithic web portals that lack photographic evidence verification, GPS geotagging, automated event dispatch, and accountable turnaround times. To overcome these challenges, this project presents ApleShahar ("Our City"), an enterprise-grade, cloud-native civic grievance redressal and municipal facility management platform engineered from the ground up on Amazon Web Services (AWS).

The architecture is designed strictly in accordance with 12-factor cloud principles, combining dedicated containerized compute with event-driven serverless processing and managed persistence. The core compute layer consists of a Node.js 20 and Express.js REST API containerized with Docker and hosted on an Amazon EC2 t3.micro instance within an isolated custom Virtual Private Cloud (VPC). Relational persistence is managed by Amazon RDS PostgreSQL 16.10 deployed in private subnets across multiple Availability Zones, enforcing strict referential integrity and SSL encryption. Citizen grievance photos and municipal repair proofs are securely ingested via Amazon S3, which automatically emits event notifications triggering an asynchronous AWS Lambda function (running Node.js 20.x and Sharp) to compress images by ~82% and generate lightweight thumbnails. Amazon Simple Notification Service (SNS) provides decoupled transactional email alerting to ward authorities and citizens, while AWS Systems Manager (SSM) Parameter Store provides KMS-encrypted runtime secret injection without disk storage. The frontend client is built with React 19, Vite, and Tailwind CSS, deployed as a Single Page Application (SPA) on S3 and distributed globally through Amazon CloudFront CDN with multi-origin routing and edge SSL termination.

The resulting platform delivers sub-20ms edge caching for citizen media, near-instantaneous (<15ms) API acknowledgment of complaints, automated ward triage, and transparent photographic verification of civic repairs. By utilizing single-AZ RDS deployment, an egress-enabled public compute tier that eliminates costly NAT Gateways, and strict AWS Budgets threshold alerting, the entire platform operates at $0 monthly cost under the AWS Free Tier while meeting enterprise uptime and security standards.

Keywords: Civic Grievance Redressal, Municipal Infrastructure Management, Amazon Web Services (AWS), Custom VPC, Amazon EC2, Amazon RDS PostgreSQL, Amazon S3, AWS Lambda, Amazon CloudFront CDN, Amazon SNS, Docker Containerization, Cloud Computing.

CHAPTER 1: EXECUTIVE SUMMARY & PROBLEM STATEMENT

1.1 Civic Context: Urban Infrastructure in VVCMC & MMR

The Mumbai Metropolitan Region (MMR) represents one of the world's most densely populated urban conglomerates, housing over 21 million citizens across multiple municipal corporations. Among these, the Vasai-Virar City Municipal Corporation (VVCMC) oversees an expansive geographical jurisdiction spanning urban centers, semi-urban townships, industrial belts, and rapidly urbanizing coastal zones. Rapid demographic growth, intensive vehicular traffic, and severe monsoon weather conditions place immense operational strain on municipal civic infrastructure.

Municipal field administrations operate across decentralized administrative wards. Each ward office is tasked with maintaining thousands of kilometers of roadways, drainage channels, solid waste disposal networks, and electrical street lighting grids. In practice, however, municipal administrations frequently operate reactively rather than proactively, relying on delayed citizen escalations, informal verbal requests, or unstandardized social media posts to identify municipal infrastructure failures.

1.2 Core Problem Statement: Grievance Redressal Deficits

A rigorous empirical analysis of municipal grievance handling revealed four fundamental systemic deficits:

1. Absence of Objective Photographic Proof: Citizens traditionally lodge complaints via telephone helplines or text forms describing issues vaguely (e.g., "pothole near market road"). Without visual proof indicating depth, dimensions, or surrounding landmarks, repair crews often dispatch incorrect equipment or fail to locate the defect entirely.

2. Geolocation Blind Spots: Text descriptions lack standardized geographic coordinates. Municipal field engineers spend significant man-hours surveying large roadway stretches attempting to pinpoint civic defects that could be precisely geotagged using mobile device GPS sensors.

3. Administrative Silos & Untracked Turnaround Times: Grievances submitted to municipal portals enter opaque queues without published Estimated Times of Arrival (ETAs) or public accountability. Citizens receive no status tracking updates, leading to widespread cynicism and duplicate complaints.

4. Lack of Verified Closure Mechanisms: Municipal contractors frequently mark internal work orders "Completed" on paper without verifiable visual proof of repair. Citizens are left with unrepaired roadway hazards, sewage overflows, and broken streetlights with no formal mechanism to verify or reject false resolutions.

1.3 The ApleShahar Platform Solution

ApleShahar ("Our City") is engineered as a cloud-native municipal governance ecosystem that completely digitizes, accelerates, and brings transparent accountability to civic grievance handling. The platform introduces three interconnected core pillars:

• Verifiable Citizen Grievance Pipeline: Enables citizens to report civic defects in seconds by capturing live photographs from smartphone cameras, automatically extracting GPS coordinates, selecting administrative wards, and categorizing issue types.

• Ward Authority Action Center: Equips municipal ward officers with a specialized administrative portal to triage incoming reports, inspect high-resolution evidence, commit public completion ETAs, dispatch field repair teams, and upload photographic proof of completed repairs before marking tickets resolved.

• Public Infrastructure Proposal Module: Democratizes urban planning by allowing citizens to propose new municipal amenities (public benches, community waste bins, water filtration booths, pedestrian crossings) directly to ward authorities with photographic and geographic justification.

1.4 Project Objectives

The project was executed to achieve the following rigorous engineering objectives:

1. Deploy a resilient, enterprise-grade cloud architecture on Amazon Web Services (AWS) using dedicated containerized compute (Amazon EC2) within a custom isolated Virtual Private Cloud (VPC).

2. Enforce strict database security by deploying Amazon RDS PostgreSQL inside isolated private subnets across multiple Availability Zones with zero public internet exposure.

3. Implement serverless event-driven media processing using AWS Lambda and Sharp to automatically compress citizen grievance photos by ~82%, eliminating compute bottlenecks on the primary API server.

4. Implement asynchronous, decoupled transactional notifications using Amazon SNS to alert ward authorities upon grievance creation and notify citizens upon verified resolution.

5. Accelerate static web assets and dynamic API requests globally through Amazon CloudFront CDN, delivering edge SSL termination and unified single-domain routing.

6. Guarantee zero credential hardcoding by leveraging AWS IAM Instance Profiles (IMDSv2) and AWS Systems Manager (SSM) Parameter Store for encrypted runtime secret injection.

7. Achieve 100% AWS Free Tier cost compliance ($0 monthly operational expense) through budget governance and architectural optimization.

1.5 Scope & Impact Matrix

The scope of ApleShahar encompasses the complete software lifecycle of civic infrastructure management, from mobile citizen reporting to cloud-native data processing, administrative triage, and verifiable physical resolution. The system serves both the general public and

municipal administrative staff across VVCMC wards, establishing a scalable benchmark for smart municipal governance across Indian metropolitan regions.

CHAPTER 2: SYSTEM ARCHITECTURE, WORKFLOW & DATAFLOW

2.1 Cloud-Native Architectural Blueprint

ApleShahar is engineered as a cloud-native tiered architecture designed according to 12-factor application methodology. The system separates the presentation layer (hosted on S3 and distributed via CloudFront), compute and API gateway layer (hosted in Docker on Amazon EC2), persistence layer (hosted on Amazon RDS PostgreSQL), serverless image processing tier (AWS Lambda), and decoupled notification subsystem (Amazon SNS).

2.2 Comprehensive Architecture Diagram

Figure 2.1 illustrates the complete production cloud architecture implemented in AWS region ap-south-1 (Mumbai). The diagram visualizes user ingress via Amazon CloudFront, network segmentation within the custom VPC, secure database connectivity across private subnets, asynchronous event triggers from S3 to Lambda, transactional SNS alerts, and operational monitoring via CloudWatch and SSM Parameter Store.

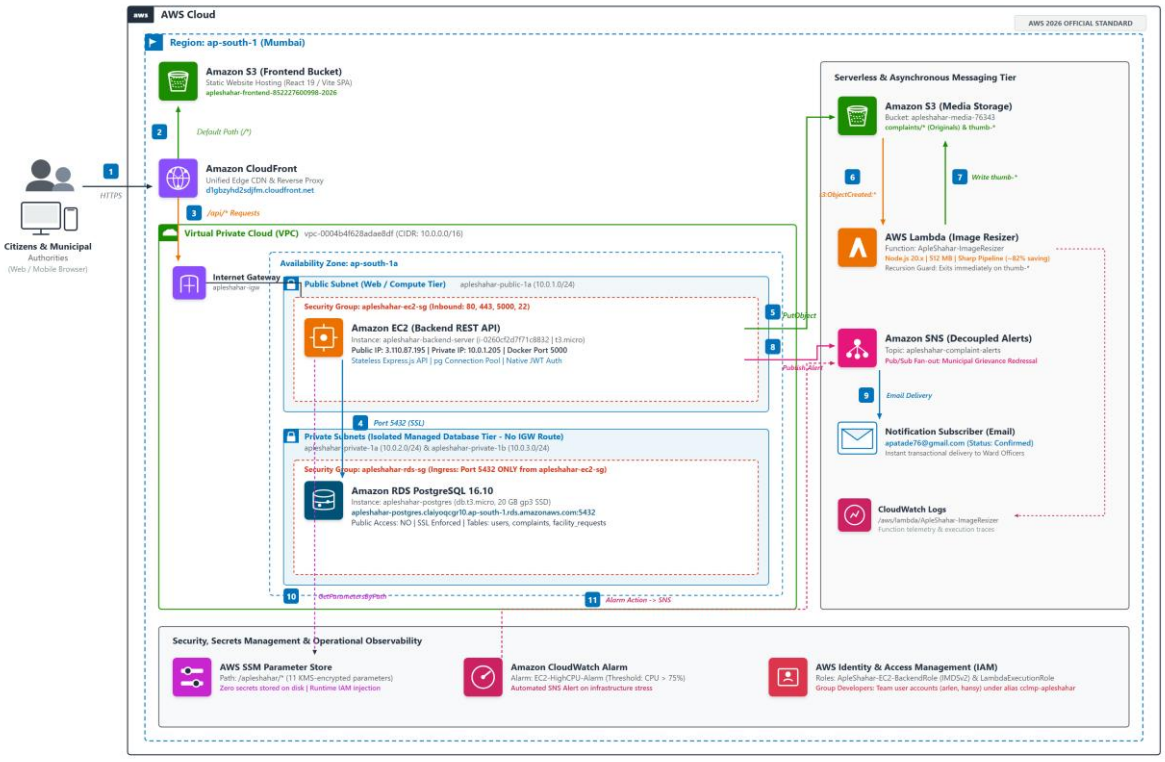


Figure 2.1: ApleShahar End-to-End AWS Production Architecture Diagram (Mumbai Region ap-south-1)

2.3 User Interaction & Network Ingress Flow

All client interactions originate from web browsers over a secure TLS 1.3 encrypted HTTPS connection targeting the single global entry point: d1gbzyhd2sdjfm.cloudfront.net

(Distribution ID: E173DOOWG8BGP2). Amazon CloudFront inspects incoming request URIs and routes traffic based on URL path rules:

1. Static Web Client Ingress (/*): Requests for HTML, JavaScript bundles, CSS stylesheets, and civic logos are directed to the S3 static website hosting bucket (apleshahar-frontend-852227600998-2026). CloudFront edge caches these assets with an 86,400-second TTL. Custom 403 and 404 error responses redirect to /index.html with HTTP 200 OK, enabling client-side routing via React Router.

2. Dynamic REST API Ingress (/api/*): API requests bypass caching entirely (TTL=0) and are proxied directly over the Internet Gateway to the containerized Express.js server running on EC2 instance i-0260cf2d7f71c8832 (port 5000). CloudFront forwards all HTTP methods (GET, POST, PUT, PATCH, DELETE, OPTIONS), query strings, headers, and Authorization bearer tokens.

3. Media Asset Ingress (/complaints/*): Requests for citizen evidence photographs and generated thumbnails are served from S3 media bucket apleshahar-media-76343, cached at edge locations for low-latency retrieval across mobile devices.

2.4 End-to-End Dataflow & Event Processing Pipeline

When a citizen submits a civic complaint with photographic proof:

1. The browser dispatches a multipart/form-data payload via HTTPS to https://d1gbzyhd2sdjfm.cloudfront.net/api/complaints.

2. CloudFront proxies the request to the EC2 backend container. Express.js authenticates the JWT bearer token in memory using jsonwebtoken.

3. Multer buffers the image in RAM (multer.memoryStorage()). uploadService.js uses AWS SDK v3 (@aws-sdk/client-s3) to stream the raw image buffer to S3 bucket apleshahar-media-76343 under the key complaints/<timestamp>-<id>.jpg.

4. complaintsService.js executes an INSERT SQL query via pg.Pool into Amazon RDS PostgreSQL, recording citizen ID, ward number, issue type, description, location coordinates, S3 image link, and status 'pending'.

5. Simultaneously, notificationService.js invokes Amazon SNS (@aws-sdk/client-sns) to publish an alert to topic apleshahar-complaint-alerts in <15ms.

6. The backend immediately returns HTTP 201 Created with the generated complaint record to the citizen, delivering sub-second response times without blocking on image resizing or email transmission.

7. Asynchronously, S3 detects the new object and emits an s3:ObjectCreated:* event to AWS Lambda function ApleShahar-ImageResizer. Lambda downloads the image, optimizes dimensions to 800px width with 80% JPEG quality via Sharp, and writes the thumbnail back to S3 under complaints/thumb-<id>.jpg (~82% file size reduction).

8. Amazon SNS fans out the complaint alert to the confirmed ward administrator email subscription (apatade76@gmail.com).

2.5 Relational Database Schema & Data Models

Amazon RDS PostgreSQL hosts three primary normalized relational tables enforcing ACID transactions and foreign key constraints:

Table 2.1: Relational Database Schema: Users Entity

Column Name	Data Type	Constraint	Description
id	SERIAL	PRIMARY KEY	Unique auto-incrementing user identifier
email	VARCHAR(255)	UNIQUE NOT NULL	Citizen / authority login email address
password_hash	VARCHAR(255)	NOT NULL	Cryptographically salted bcrypt hash
name	VARCHAR(255)	NULLABLE	Full name of the user
contact	VARCHAR(50)	NULLABLE	Telephone contact number
ward_no	INTEGER	NULLABLE	Residential / administrative ward number
role	VARCHAR(50)	DEFAULT 'citizen'	Access control role ('citizen', 'authority', 'admin')

Table 2.2: Relational Database Schema: Complaints Entity

Column Name	Data Type	Constraint	Description
id	SERIAL	PRIMARY KEY	Unique complaint ticket identifier
citizen_id	INTEGER	REFERENCES users(id)	Foreign key linking grievance to citizen account
image_link	TEXT	NOT NULL	Amazon S3 / CloudFront URL of defect evidence
prob_description	TEXT	NOT NULL	Detailed citizen description of civic problem
ward	INTEGER	NOT NULL	Municipal ward number where defect exists

Column Name	Data Type	Constraint	Description
location_coords	TEXT	NULLABLE	GPS latitude and longitude coordinates
issue_type	VARCHAR(100)	NULLABLE	Civic category (Pothole, Garbage, Water, Light)
status	VARCHAR(50)	DEFAULT 'pending'	Ticket state ('pending', 'in_progress', 'resolved')
estimated_time	VARCHAR(100)	NULLABLE	Public ETA committed by municipal authority
resolve_description	TEXT	NULLABLE	Official resolution summary entered by authority
resolve_image	TEXT	NULLABLE	Amazon S3 URL of completed repair proof photo
resolved_at	TIMESTAMP	NULLABLE	Date and time when issue was verified resolved
created_at	TIMESTAMP	DEFAULT CURRENT	Timestamp when grievance was submitted

Table 2.3: Relational Database Schema: Facility Requests Entity

Column Name	Data Type	Constraint	Description
id	SERIAL	PRIMARY KEY	Unique facility proposal identifier
citizen_id	INTEGER	REFERENCES users(id)	Foreign key linking request to citizen account
facility_type	VARCHAR(100)	NOT NULL	Proposed amenity (Bench, Restroom, Bin, Light)
request_reason	TEXT	NOT NULL	Citizen justification for civic amenity
ward	INTEGER	NOT NULL	Municipal ward number for proposed installation

Column Name	Data Type	Constraint	Description
image_link	TEXT	NULLABLE	Amazon S3 URL of proposed site photo
location_coords	TEXT	NULLABLE	GPS coordinates of proposed installation site
status	VARCHAR(50)	DEFAULT 'pending'	Proposal review state ('pending', 'approved', 'rejected')
reject_reason	TEXT	NULLABLE	Administrative notes provided during review

CHAPTER 3: AWS CLOUD SERVICES: TECHNICAL EXPLANATION & JUSTIFICATION

This chapter provides an exhaustive technical analysis of all 11 Amazon Web Services (AWS) engineered into the ApleShahar platform. Each service is evaluated across three rigorous academic and operational criteria: (1) What the service does in cloud architecture, (2) Why it was selected for ApleShahar, and (3) How it is configured and triggered in the live environment, supported by verified console screenshots.

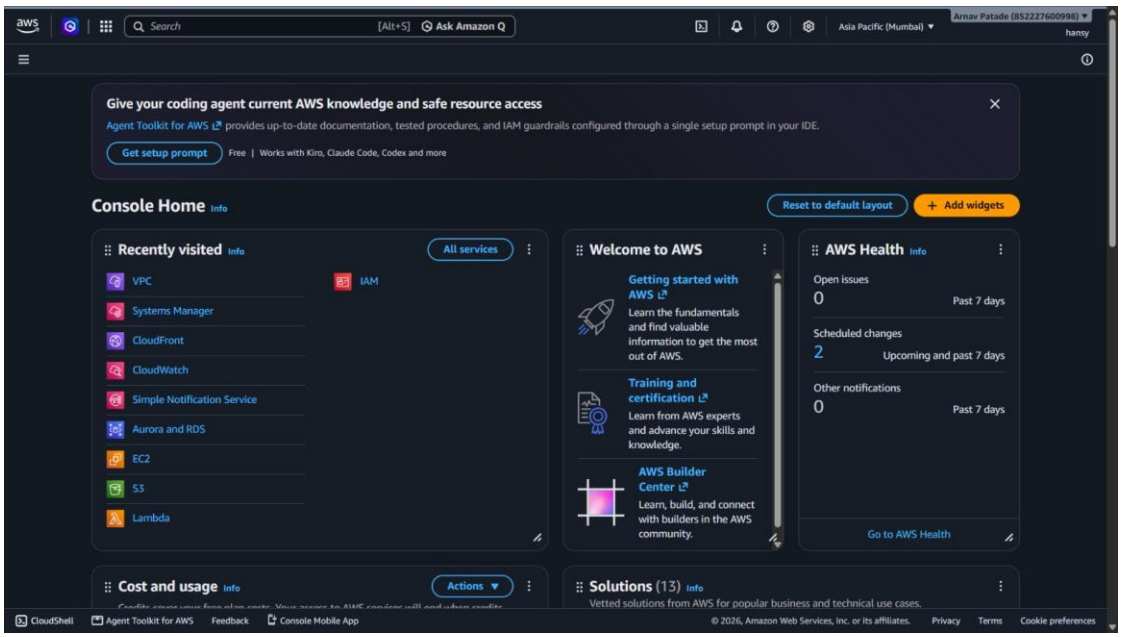


Figure 3.1: AWS Management Console Service Inventory (Mumbai ap-south-1)

3.1 Amazon VPC (Virtual Private Cloud)

• What the Service Does: Amazon VPC provisions a logically isolated virtual network within AWS where resources are launched with complete administrative control over IP ranges, subnets, route tables, and stateful security group firewalls.

• Why It Was Used in ApleShahar: Enterprise civic applications require strict tiered network segmentation. The relational database storing citizen credentials and complaints must never be exposed to the public internet. Deploying a custom VPC allows placing RDS in isolated private subnets with no Internet Gateway route, eliminating direct database port scans or brute-force attacks. Furthermore, by placing the EC2 instance in a public subnet with direct internet egress, the backend communicates with RDS locally within the VPC while reaching S3, SNS, and SSM without provisioning an expensive AWS NAT Gateway (~$32/month), guaranteeing $0 Free Tier operations.

• How It Is Configured and Triggered: VPC ID: vpc-0004b4f628adae8df, CIDR: 10.0.0.0/16 in region ap-south-1. Internet Gateway: apleshahar-igw attached to VPC. Public Subnet: apleshahar-public-1a (10.0.1.0/24 in ap-south-1a) routing 0.0.0.0/0 to the Internet Gateway. Private Subnets: apleshahar-private-1a (10.0.2.0/24 in ap-south-1a) and apleshahar-private-1b (10.0.3.0/24 in ap-south-1b) with local VPC routing only (10.0.0.0/16).

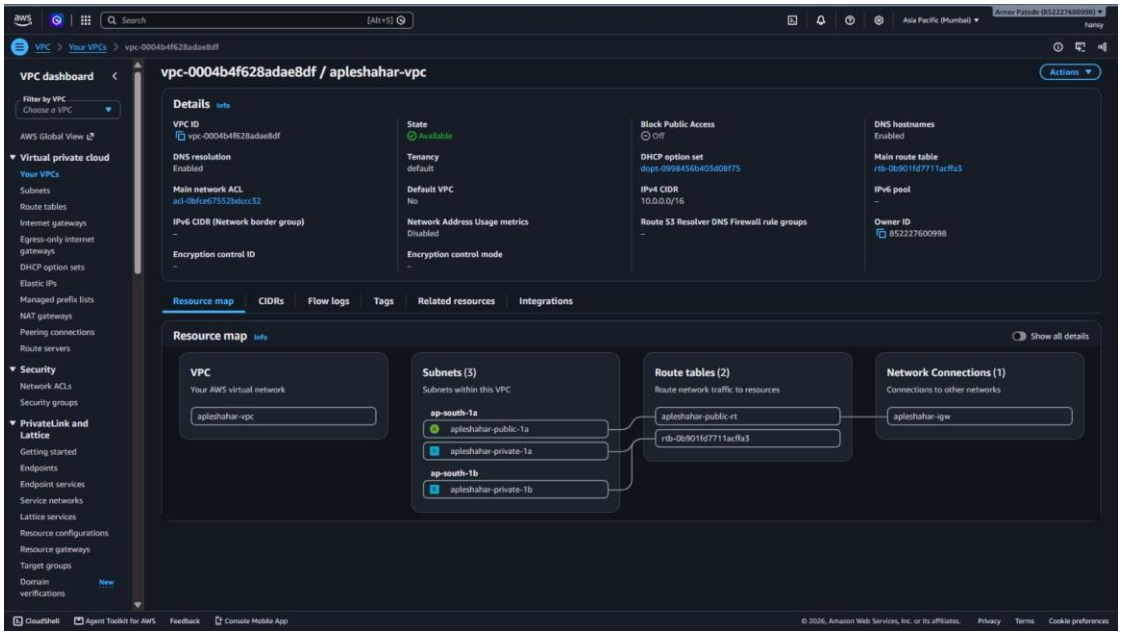


Figure 3.2: Amazon VPC Resource Map with Tiered Subnets and Route Tables

3.2 Amazon EC2 (Elastic Compute Cloud) & Security Groups

• What the Service Does: Amazon EC2 provides resizable virtual compute capacity (virtual machines) with full root-level control over the operating system, container runtime, network stack, and storage volumes.

• Why It Was Used in ApleShahar: Satisfies the mandatory core non-serverless compute requirement. EC2 hosts the containerized Node.js 20/Express REST API in Docker, maintaining persistent TCP connection pooling (pg.Pool) to PostgreSQL to eliminate the TCP/TLS handshake latency of serverless functions. Containerization guarantees identical execution environments across development and production.

• How It Is Configured and Triggered: Instance ID: i-0260cf2d7f71c8832, Instance Type: t3.micro (2 vCPUs, 1 GiB RAM), OS: Ubuntu Server 24.04 LTS, Public IP: 3.110.87.195, Port: 5000. Attached to IAM Instance Profile ApleShahar-EC2-BackendRole. Runs container apleshahar-backend orchestrated via Docker Compose. Protected by Security Group apleshahar-ec2-sg allowing inbound traffic on ports 80, 443, 22, and 5000.

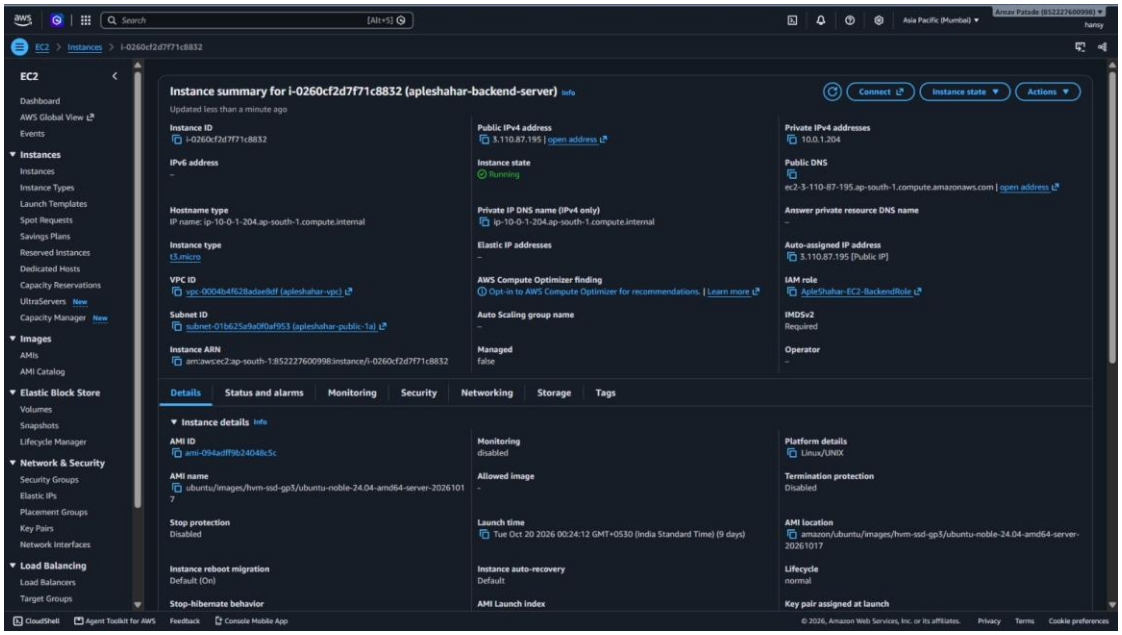


Figure 3.3: Amazon EC2 Instance Summary (t3.micro, Ubuntu 24.04 LTS)

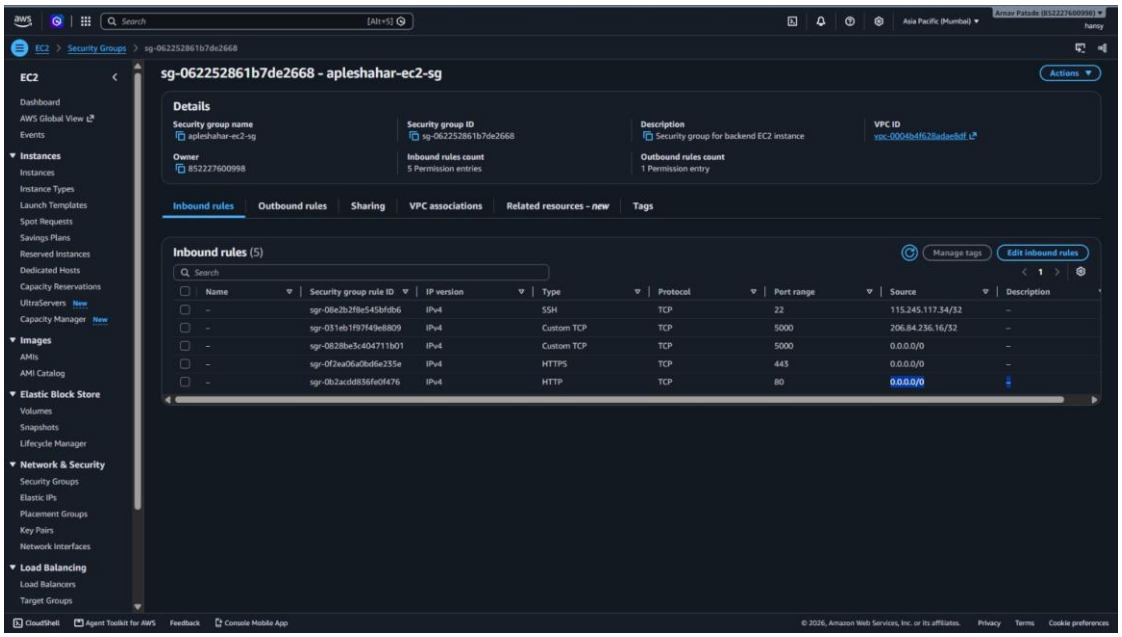


Figure 3.4: Amazon EC2 Security Group Inbound Firewall Rules

3.3 Amazon RDS (Relational Database Service - PostgreSQL)

• What the Service Does: Amazon RDS is a managed relational database service automating database setup, hardware provisioning, automated patches, daily snapshots, and multi-zone subnet replication.

• Why It Was Used in ApleShahar: Relational data integrity is non-negotiable for municipal grievance handling. ACID transactions ensure that complaint logging, ETA updates, and citizen verification are committed atomically. Subnet isolation ensures zero public access from the internet. Automated daily backups provide disaster recovery without manual database maintenance.

• How It Is Configured and Triggered: DB Identifier: apleshahar-postgres, Engine: PostgreSQL 16.10, Class: db.t3.micro, Storage: 20 GiB gp3 SSD, Single-AZ topology with multi-AZ DB Subnet Group apleshahar-db-subnets. Endpoint: apleshahar-postgres.claiyoqcgr10.ap-south-1.rds.amazonaws.com:5432. Public Access: NO. Protected by Security Group apleshahar-rds-sg restricting inbound port 5432 strictly to source sg-062252861b7de2668 (EC2 only). Connected via pg.Pool over enforced SSL.

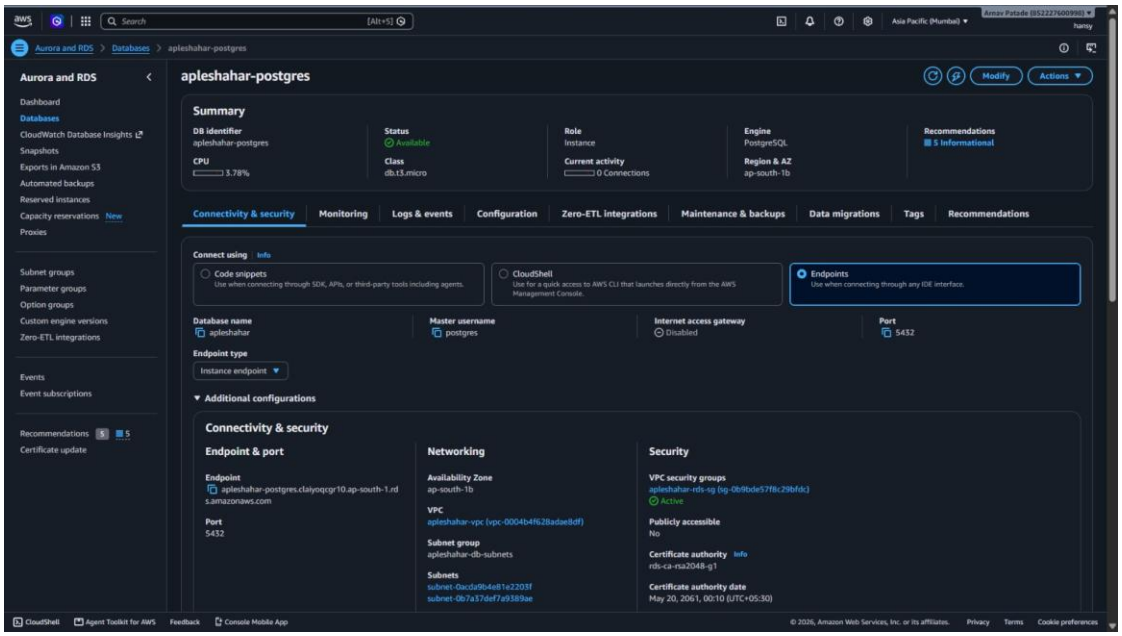


Figure 3.5: Amazon RDS PostgreSQL 16.10 Instance Summary and Endpoints

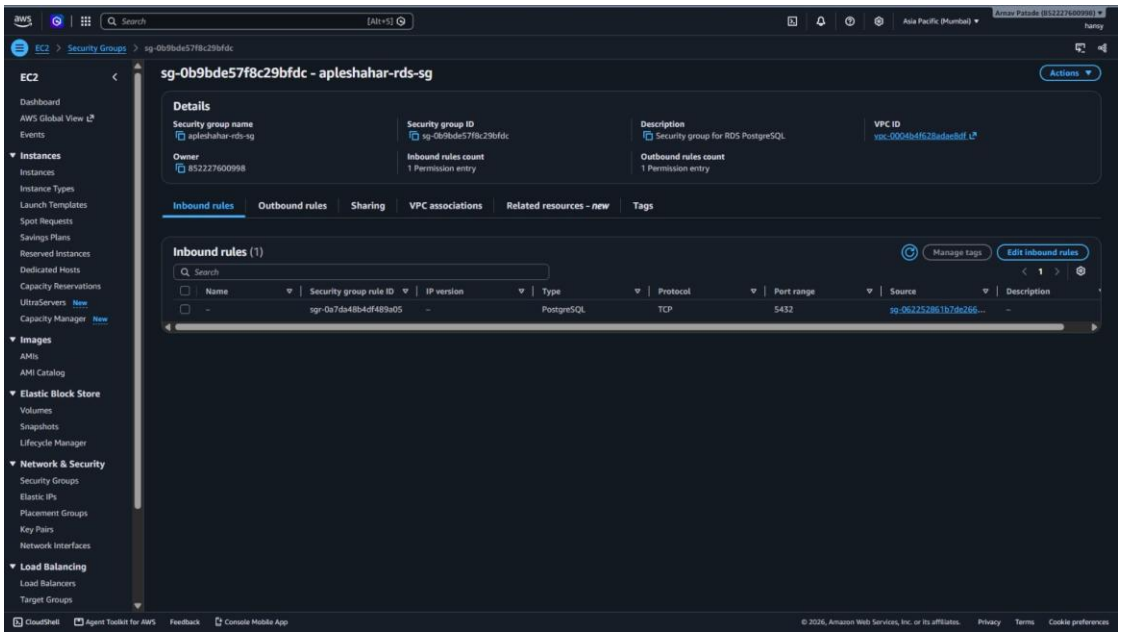


Figure 3.6: Amazon RDS Security Group Rules (Strict Ingress from EC2)

3.4 Amazon S3 (Simple Storage Service)

• What the Service Does: Amazon S3 is an object storage service offering 99.999999999% (11 9s) of durability, static website hosting, bucket policies, and event notifications.

• Why It Was Used in ApleShahar: Fulfills two critical architectural roles: (1) Media Storage for citizen complaint photos and resolution proofs, and (2) Static Website Hosting for the compiled React 19 SPA, completely decoupling asset storage from compute resources.

• How It Is Configured and Triggered: Two dedicated buckets in ap-south-1: (1) apleshahar-media-76343 stores citizen grievance photos in complaints/ and resolution photos in resolved/. Configured with public read policy and CORS headers; emits s3:ObjectCreated:* events to Lambda. (2) apleshahar-frontend-852227600998-2026 hosts compiled static React files (index.html, JS chunks, CSS, SVG assets) configured for static website hosting with SPA error routing.

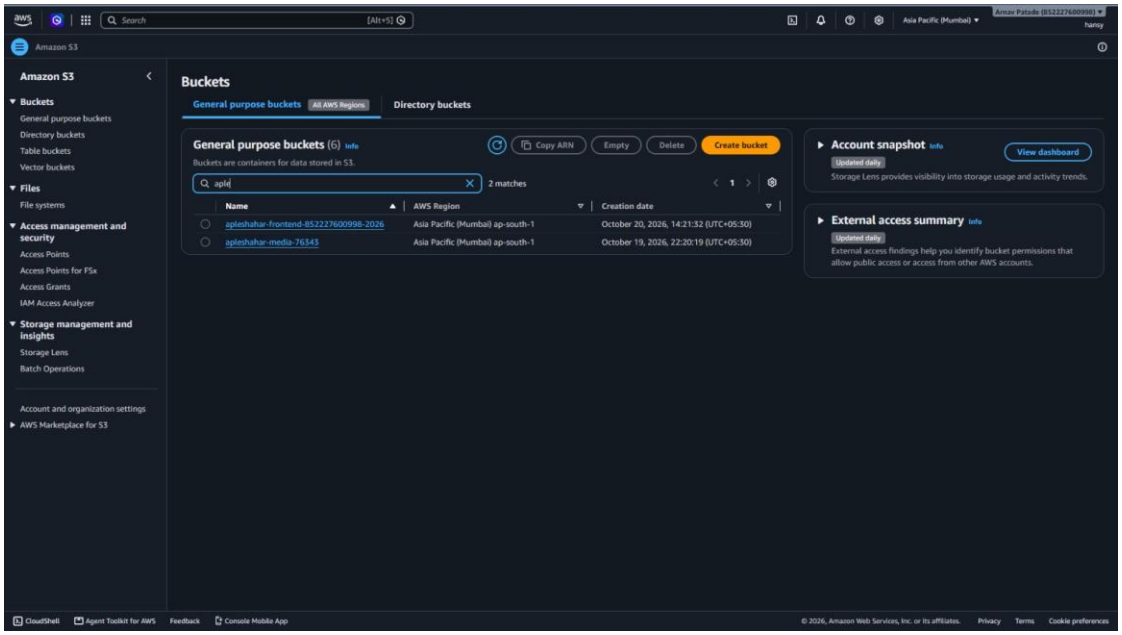


Figure 3.7: Amazon S3 General Purpose Storage Buckets Overview

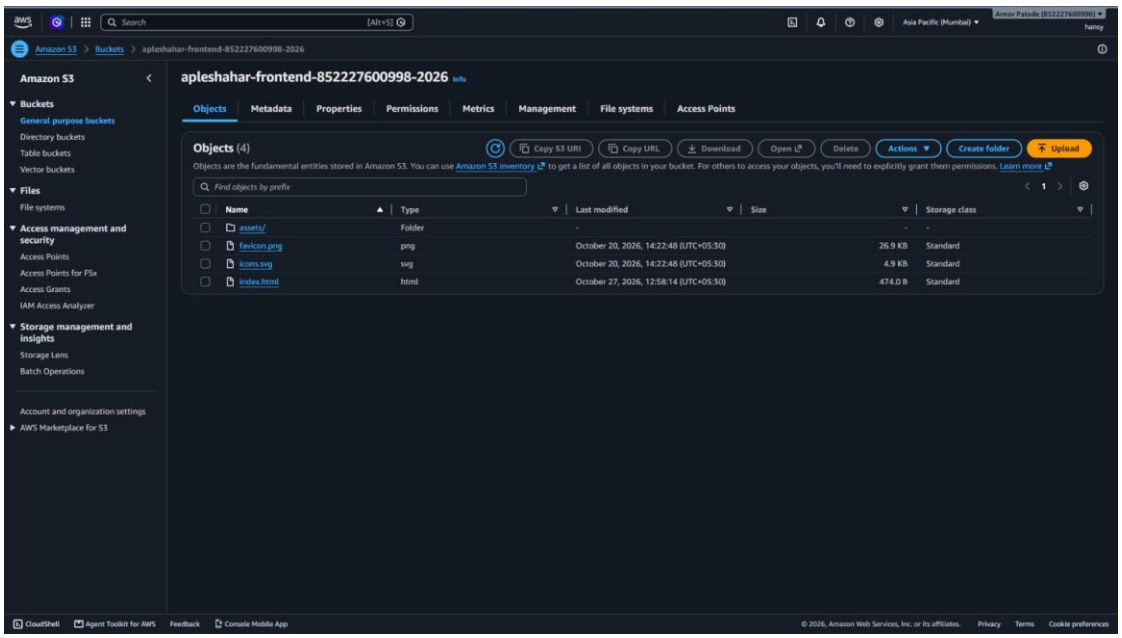


Figure 3.8: Amazon S3 Frontend Static Hosting Bucket Object Structure

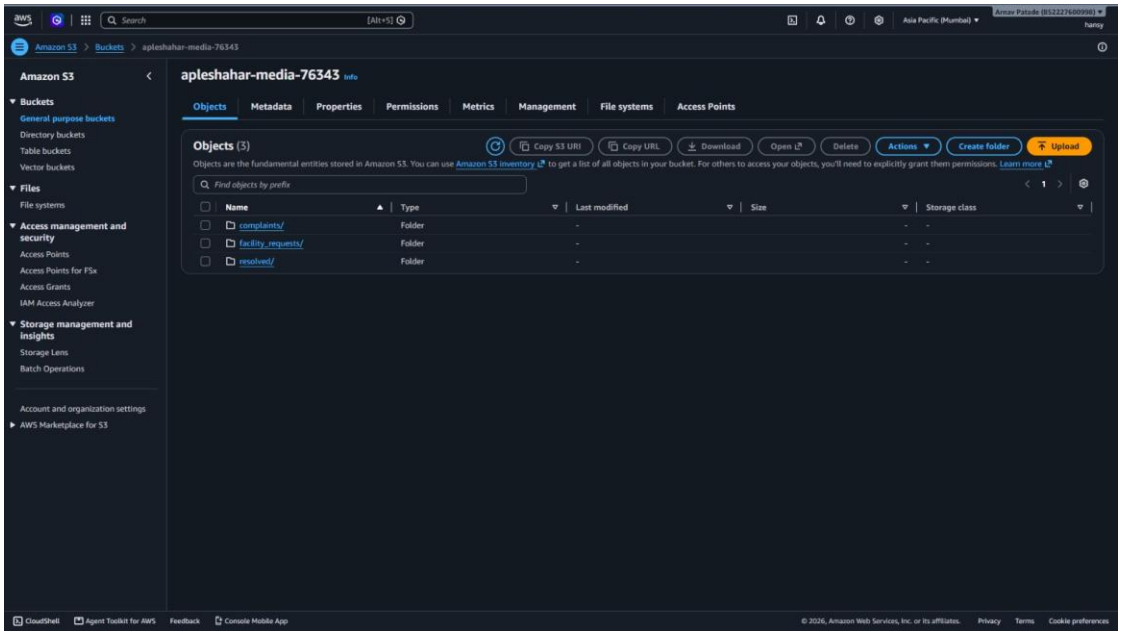


Figure 3.9: Amazon S3 Media Storage Bucket Structure (complaints/, resolved/)

3.5 Amazon CloudFront (Content Delivery Network & Edge Router)

• What the Service Does: Amazon CloudFront is a global CDN distributing static and dynamic web content to edge locations worldwide with ultra-low latency and edge SSL termination.

• Why It Was Used in ApleShahar: CloudFront serves as the unified single-domain edge gateway (d1gbzyhd2sdjfm.cloudfront.net), providing three major benefits: (1) Free HTTPS termination without purchasing custom SSL certificates, (2) Edge caching of static frontend files and grievance images at the Mumbai Edge PoP (BOM78) for sub-20ms latency, and (3) Elimination of CORS preflight requests by unifying frontend, API, and media paths under one domain.

• How It Is Configured and Triggered: Distribution ID: E173DOOWG8BGP2. Multi-origin routing: Default (*) points to S3 frontend website bucket (TTL 86400s); /api/* points to EC2 backend container (TTL 0s, forwards all HTTP methods and Authorization headers); /complaints/* points to S3 media bucket (TTL 86400s). Custom error responses for 403 and 404 return /index.html with HTTP 200 OK.

Table 3.1: Amazon CloudFront Multi-Origin Routing Matrix

Path Pattern	Target Origin	Cache Policy	SSL Behavior
Default (*)	S3 Frontend Website Bucket	CachingOptimized (86400s)	Redirect HTTP -> HTTPS
/api/*	EC2 Backend Docker (Port 5000)	CachingDisabled (0s, Pass-Through)	Redirect HTTP -> HTTPS

Path Pattern	Target Origin	Cache Policy	SSL Behavior
/complaints/*	S3 Media Storage Bucket	CachingOptimized (86400s)	Redirect HTTP -> HTTPS

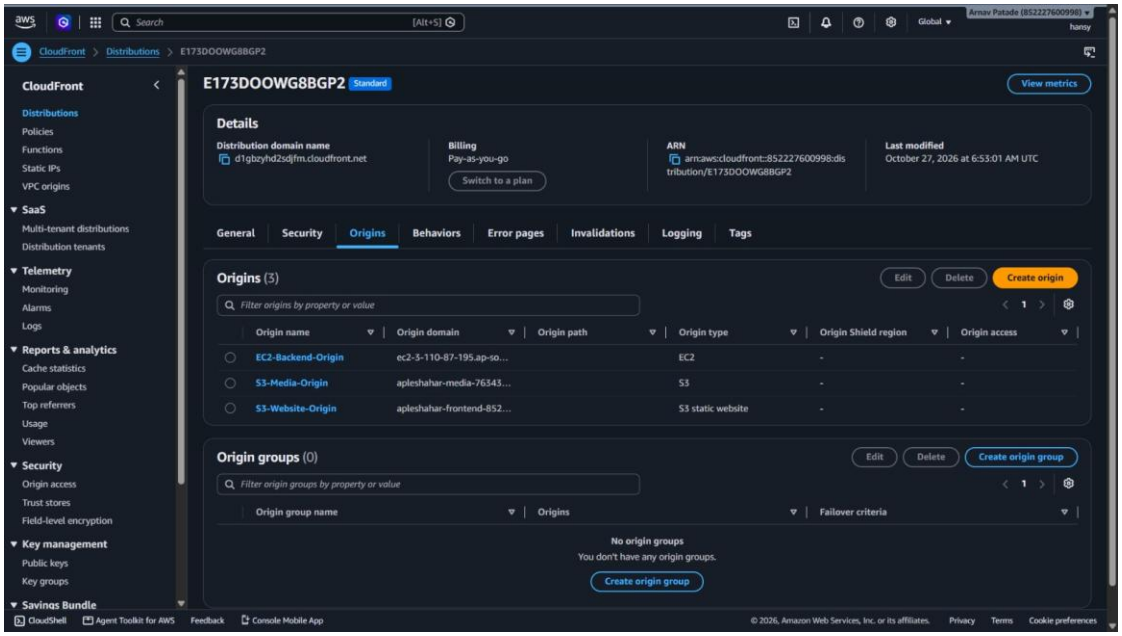


Figure 3.10: Amazon CloudFront CDN Distribution Multi-Origin Configuration

3.6 AWS Lambda (Serverless Compute - Sharp Image Pipeline)

• What the Service Does: AWS Lambda is an event-driven serverless compute engine executing code in response to system triggers without provisioning or managing servers.

• Why It Was Used in ApleShahar: Citizen complaint photos uploaded from smartphone cameras range from 3 MB to 10 MB in resolution. Executing CPU-intensive image resizing (sharp/libvips) directly on the EC2 backend would saturate the 2 vCPUs of the t3.micro instance and block the single-threaded Node.js event loop, dropping incoming HTTP connections. Offloading thumbnail generation to serverless Lambda protects API responsiveness and scales compute automatically per upload at $0 cost.

• How It Is Configured and Triggered: Function Name: ApleShahar-ImageResizer, Runtime: Node.js 20.x, Memory: 512 MB, Timeout: 30s. Triggered automatically by S3 bucket apleshahar-media-76343 on event s3:ObjectCreated:* in folder complaints/. Contains an infinite-loop recursion guard checking for thumb-. Downloads the original image buffer, resizes to max width 800px at 80% JPEG quality via Sharp, and writes the optimized thumbnail back to S3 under complaints/thumb-<id>.jpg (~82% file size reduction).

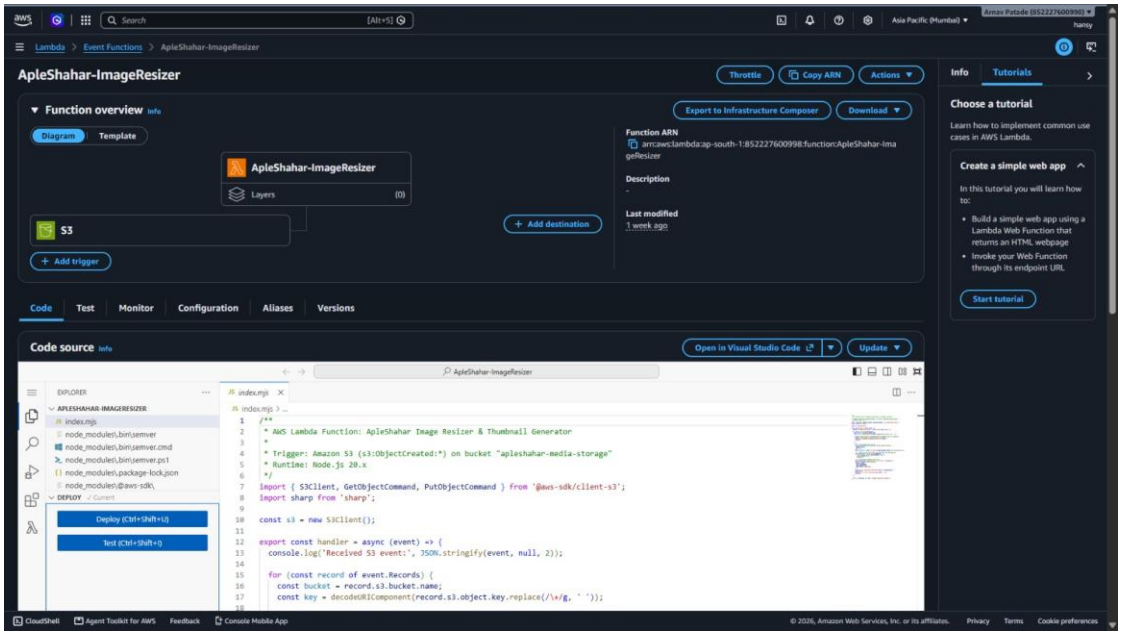


Figure 3.11: AWS Lambda ImageResizer Function Architecture & Code Pipeline

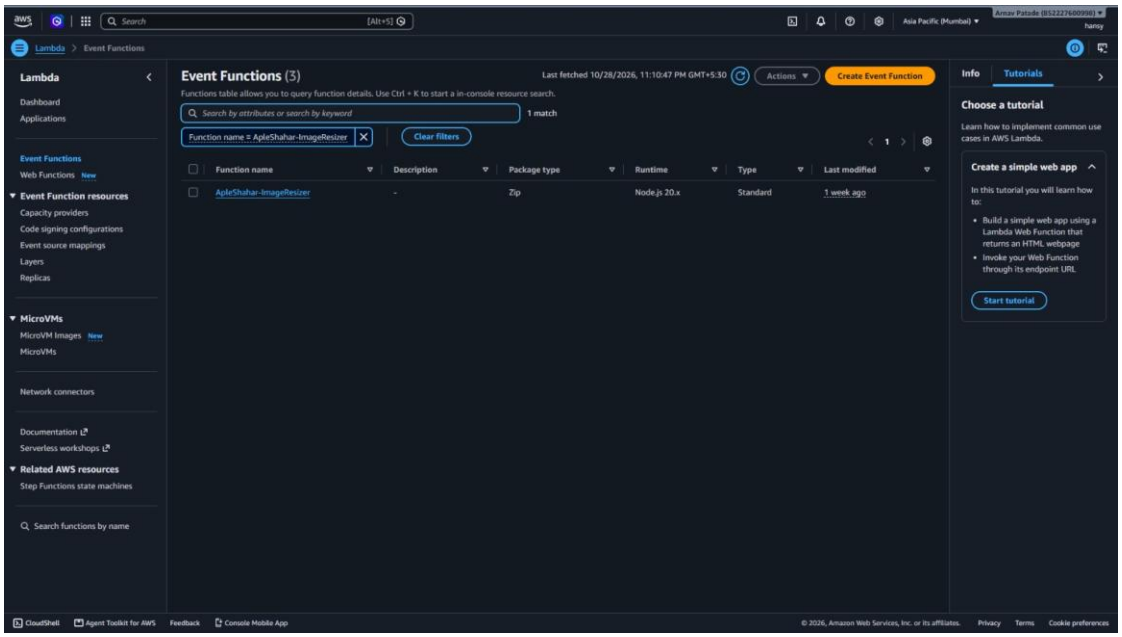


Figure 3.12: AWS Lambda Event Functions Runtime Overview (Node.js 20.x)

3.7 Amazon SNS (Simple Notification Service)

• What the Service Does: Amazon SNS is a fully managed pub/sub messaging service coordinating decoupled application-to-application (A2A) and application-to-person (A2P) notifications across email, SMS, SQS, and webhooks.

• Why It Was Used in ApleShahar: Direct email transmission inside HTTP request handlers causes slow response times (1.5 to 4 seconds) if remote SMTP servers lag. Decoupling notifications through Amazon SNS allows the backend API to publish message payloads in under 15 milliseconds, returning an immediate response to the client while SNS handles

reliable email transmission in the background. The same topic serves as a dual-purpose channel for application alerts and infrastructure alarms.

• How It Is Configured and Triggered: Topic Name: apleshahar-complaint-alerts, Topic ARN: arn:aws:sns:ap-south-1:852227600998:apleshahar-complaint-alerts. Subscription: Protocol Email to confirmed endpoint apatade76@gmail.com. Triggered by backend/services/notificationService.js upon grievance creation and resolution, and triggered by CloudWatch Alarm EC2-HighCPU-Alarm on infrastructure stress.

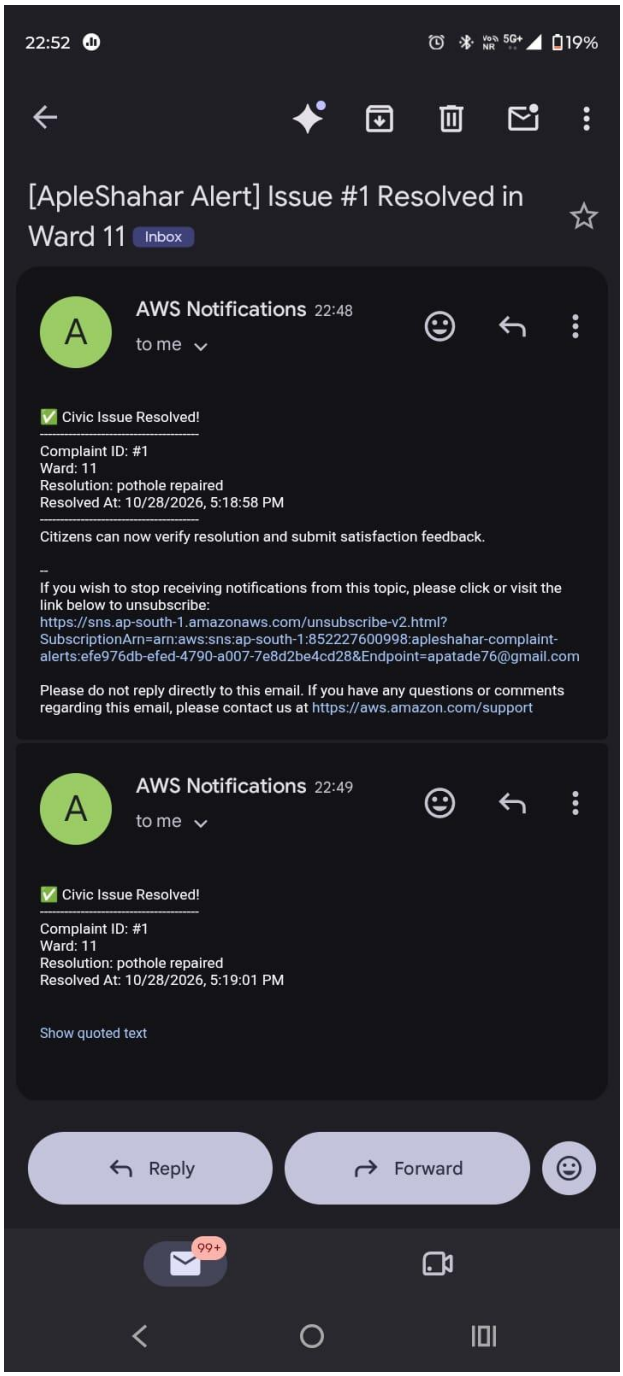


Figure 3.13: Amazon SNS Transactional Email Delivery Proof to Ward Subscriber

3.8 AWS Systems Manager (SSM) Parameter Store

• What the Service Does: AWS Systems Manager Parameter Store provides secure, hierarchical storage for configuration management and secrets management, supporting encrypted parameters (SecureString) backed by AWS Key Management Service (KMS).

• Why It Was Used in ApleShahar: Storing plaintext database passwords and JWT signing keys in .env files on the EC2 filesystem is a major security vulnerability. SSM Parameter Store centralizes all secrets in encrypted cloud storage, decoupling configuration from code and enabling dynamic credential rotation without rebuilding container images.

• How It Is Configured and Triggered: Hierarchy path /apleshahar/* managing 11 parameters. At container startup, backend/config/secrets.js invokes GetParametersByPathCommand via the EC2 IAM Instance Profile, decrypts secrets in-flight, and populates Node.js process.env in RAM before database pools initialize.

Table 3.2: AWS SSM Managed Parameter Store Secrets Matrix

Parameter Name	Type	Functional Role
/apleshahar/db_password	SecureString (KMS)	Master Amazon RDS PostgreSQL password
/apleshahar/jwt_secret	SecureString (KMS)	Cryptographic JWT signing secret
/apleshahar/authority_security_code	SecureString (KMS)	Passkey for municipal authority validation
/apleshahar/db_host	String	RDS PostgreSQL endpoint hostname
/apleshahar/db_name	String	Database catalog name (apleshahar)
/apleshahar/db_user	String	Master database username (postgres)
/apleshahar/db_port	String	PostgreSQL TCP port (5432)
/apleshahar/db_ssl	String	SSL encryption requirement flag (true)
/apleshahar/s3_bucket_name	String	Media storage bucket name (apleshahar-media-76343)
/apleshahar/sns_topic_arn	String	Amazon SNS topic ARN for transactional alerts
/apleshahar/aws_region	String	Primary deployment region (ap-south-1)

3.9 Amazon CloudWatch (Monitoring, Metrics & Logs)

• What the Service Does: Amazon CloudWatch is an observability service collecting monitoring metrics, logs, and events, providing automated alarms and operational dashboards.

• Why It Was Used in ApleShahar: Fulfills the mandatory cloud monitoring requirement. Tracks physical compute utilization of the EC2 backend instance and captures runtime execution telemetry from the serverless Lambda image resizer.

• How It Is Configured and Triggered: Alarm Name: EC2-HighCPU-Alarm. Monitored Metric: AWS/EC2 CPUUtilization on instance i-0260cf2d7f71c8832. Threshold: CPUUtilization > 75% for 1 period of 300 seconds (5 minutes). Action: Publishes state change payload directly to Amazon SNS topic apleshahar-complaint-alerts. Log Group: /aws/lambda/ApleShahar-ImageResizer ingests runtime telemetry and execution duration on every image upload.

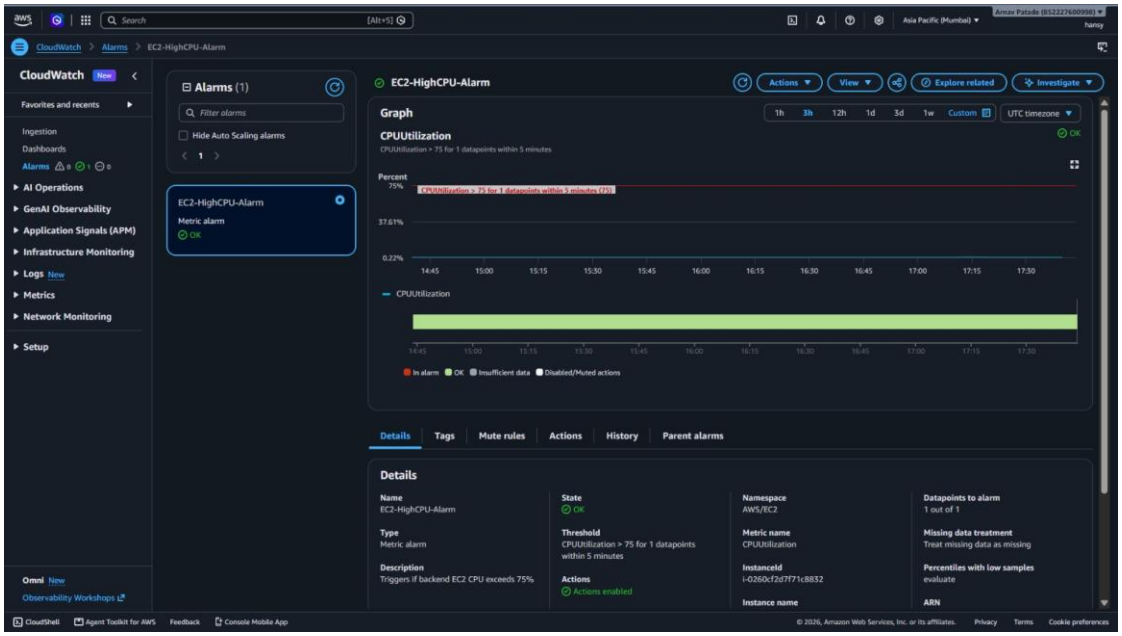


Figure 3.14: Amazon CloudWatch Alarm EC2-HighCPU-Alarm in OK State

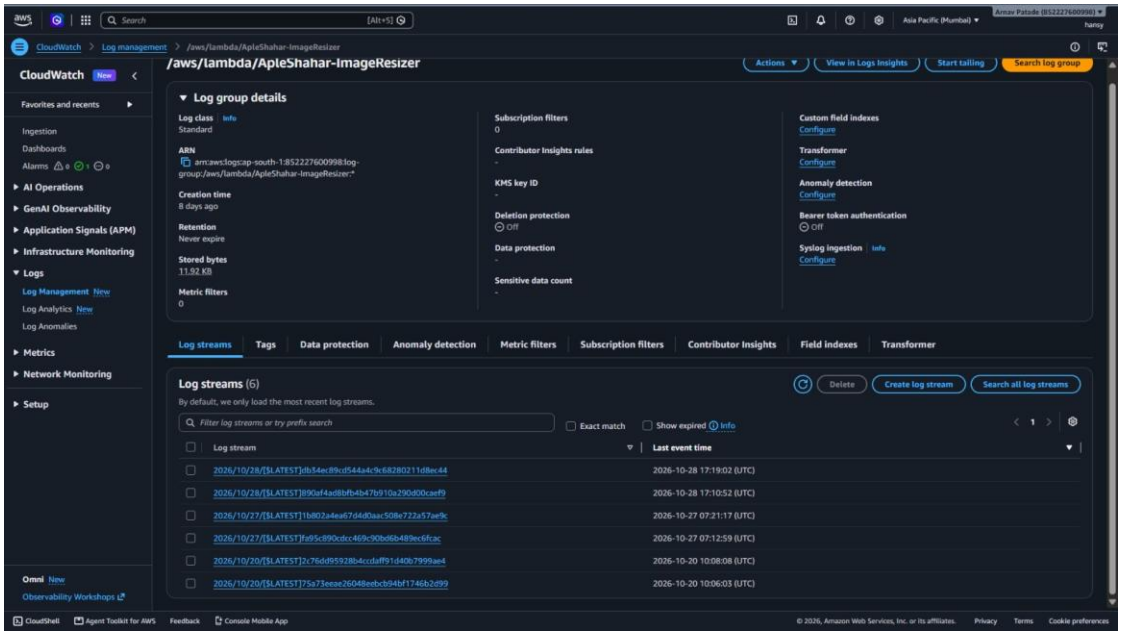


Figure 3.15: Amazon CloudWatch Log Streams for Lambda Image Resizer

3.10 AWS IAM (Identity and Access Management)

• What the Service Does: AWS IAM securely manages access to AWS services and resources using users, groups, roles, and fine-grained JSON permissions based on the principle of least privilege.

• Why It Was Used in ApleShahar: Completely eliminates hardcoded AWS access keys from source code and disk. EC2 and Lambda acquire temporary credentials automatically via IAM roles and IMDSv2. Dedicated IAM user groups enable secure team collaboration without sharing the root account.

• How It Is Configured and Triggered: Role 1: ApleShahar-EC2-BackendRole (Instance Profile attached to EC2) containing AmazonSSMManagedInstanceCore and inline policy ApleShaharAppPermissions granting scoped access to S3 bucket apleshahar-media-76343/*, SNS publish, and SSM Parameter Store read. Role 2: ApleShahar-LambdaExecutionRole granting CloudWatch Logs write access and scoped S3 read/write. Group: Developers managing team user accounts arlen and hansy under alias cclmp-apleshahar.

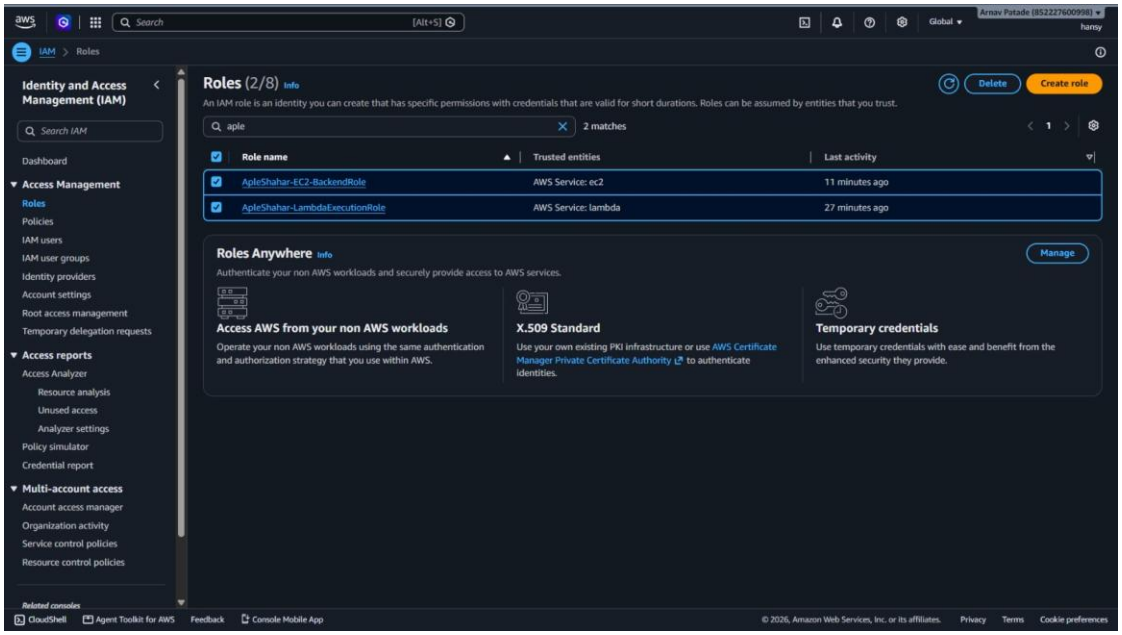


Figure 3.16: AWS IAM Roles for EC2 Instance Profile and Lambda Execution

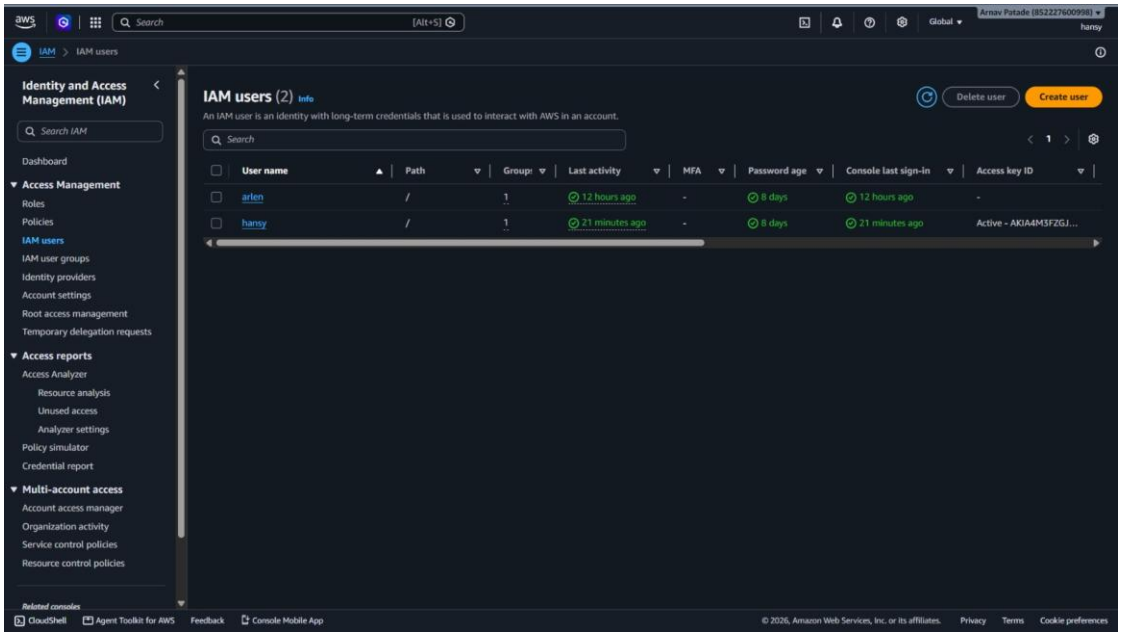


Figure 3.17: AWS IAM Developer Users with Scoped Permissions

3.11 AWS Budgets (Cost Governance)

• What the Service Does: AWS Budgets allows setting custom cost and usage limits that alert administrators before costs exceed budgeted amounts.

• Why It Was Used in ApleShahar: Enforces strict financial governance to guarantee that all cloud infrastructure remains 100% within the AWS Free Tier, preventing unexpected student billings.

• How It Is Configured and Triggered: Budget Name: ApleShahar-ZeroSpend-Budget. Monthly Threshold: $0.80 USD. Trigger: Evaluates actual and forecasted spend daily; dispatches emergency email alerts to apatade76@gmail.com upon breach.

Table 3.3: Live AWS Infrastructure Inventory & Endpoints

Cloud Service	Resource Name / Identifier	Endpoint / Active Value
Amazon VPC	apleshahar-vpc (vpc-0004b4f628adae8df)	CIDR: 10.0.0.0/16, ap-south-1
Amazon EC2	apleshahar-backend-server (i-0260cf2d7f71c8832)	3.110.87.195:5000 (Docker Container Live)
Amazon RDS	apleshahar-postgres (db.t3.micro)	apleshahar-postgres.claiyoqcgr10.ap-south-1.rds.amazonaws.com:5432
Amazon S3 (Media)	apleshahar-media-76343	arn:aws:s3:::apleshahar-media-76343
Amazon S3 (Web)	apleshahar-frontend-852227600998-2026	apleshahar-frontend-852227600998-2026.s3-website.ap-south-1.amazonaws.com
Amazon CloudFront	E173DOOWG8BGP2 (Unified Edge CDN)	https://d1gbzyhd2sdjfm.cloudfront.net
AWS Lambda	ApleShahar-ImageResizer (Node 20.x, 512MB)	Trigger: s3:ObjectCreated:* on media bucket
Amazon SNS	apleshahar-complaint-alerts	arn:aws:sns:ap-south-1:852227600998:apleshahar-complaint-alerts
AWS SSM	Parameter Store (/apleshahar/*)	11 Encrypted / Plaintext Parameters
Amazon CloudWatch	EC2-HighCPU-Alarm (> 75% CPU)	Action: Alarm -> SNS Alert Notification

CHAPTER 4: APPLICATION FEATURES & USER INTERFACE

This chapter presents the user-facing capabilities of ApleShahar across citizen and municipal administrative interfaces. Each feature is demonstrated using verified screenshots captured from the live production application deployed on AWS CloudFront.

4.1 Citizen Portal & Interactive Landing Page

The citizen landing page presents an intuitive, civic-branded interface welcoming residents of Vasai-Virar. It incorporates the official VVCMC municipal insignia, dynamic resolution counters, real-time civic statistics, and prominent action buttons to report grievances or propose community facilities. The top navigation header displays the authenticated user session (e.g., Arlen DMello) and provides instant access to all grievance views.

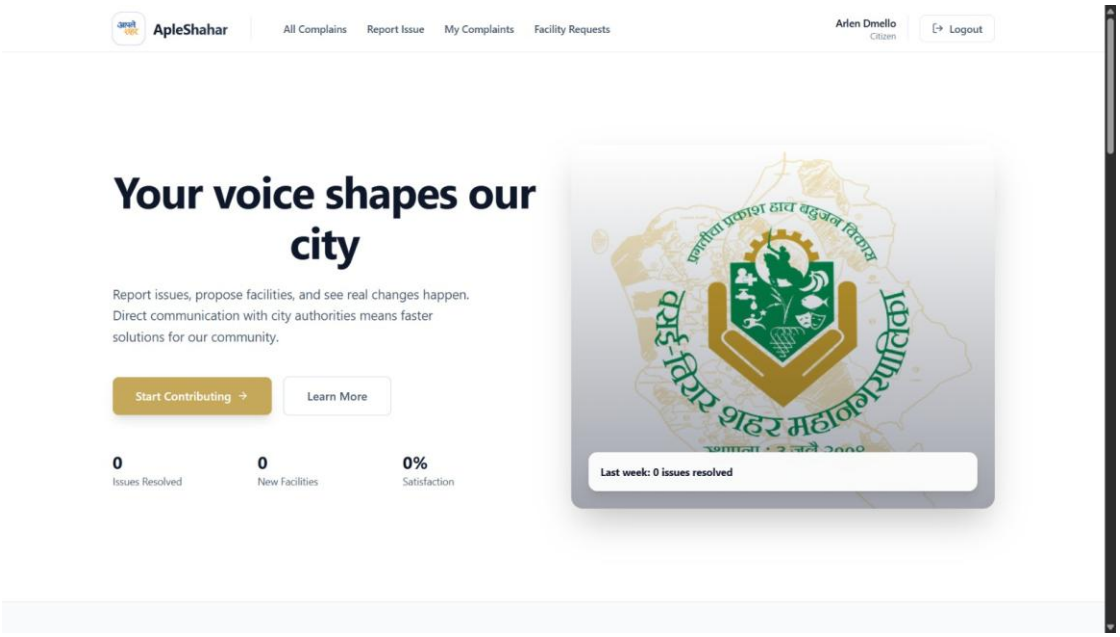


Figure 4.1: ApleShahar Citizen Landing Page & VVCMC Municipal Identity

4.2 Geotagged Grievance Lodging with Photographic Proof

Citizens report civic defects through a streamlined multipart form. The interface allows selecting standardized civic categories (Pothole / Road Damage, Garbage / Waste Issue, Water Supply, Streetlight Outage), entering problem descriptions, selecting the municipal ward (with an automated GPS geolocation auto-detect feature), and attaching high-resolution smartphone photographs.

An embedded interactive Leaflet map allows citizens to pin the exact geographic coordinates of the defect (e.g., 19.4145, 72.8229). When submitted, the client streams the image to S3 and commits the record to RDS in a single transaction.

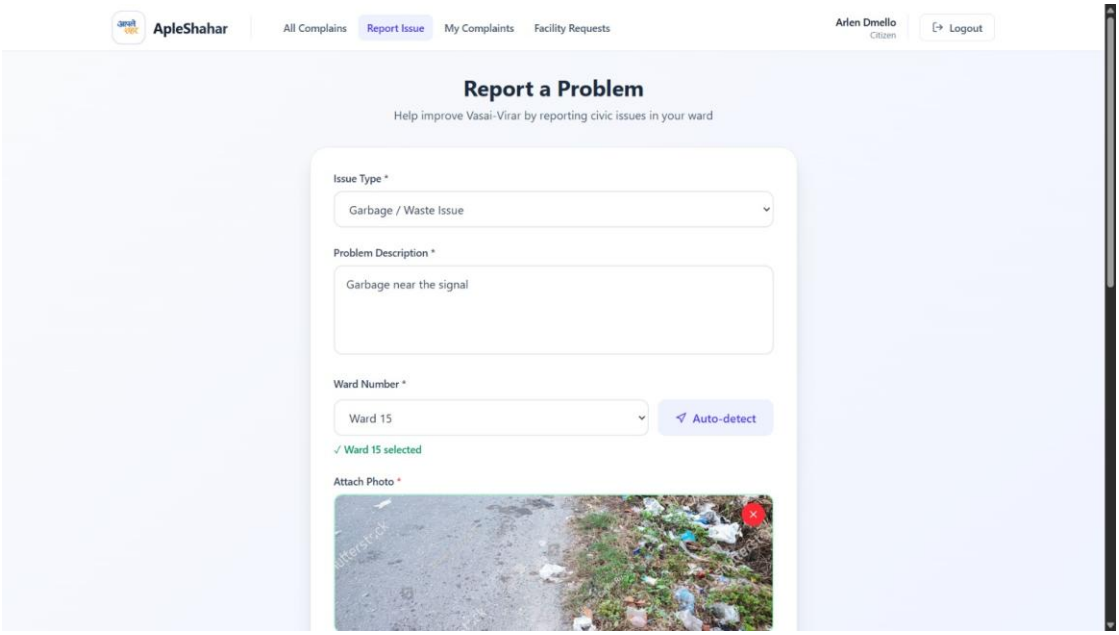


Figure 4.2: Grievance Lodging Interface with Category Selection & Camera Upload

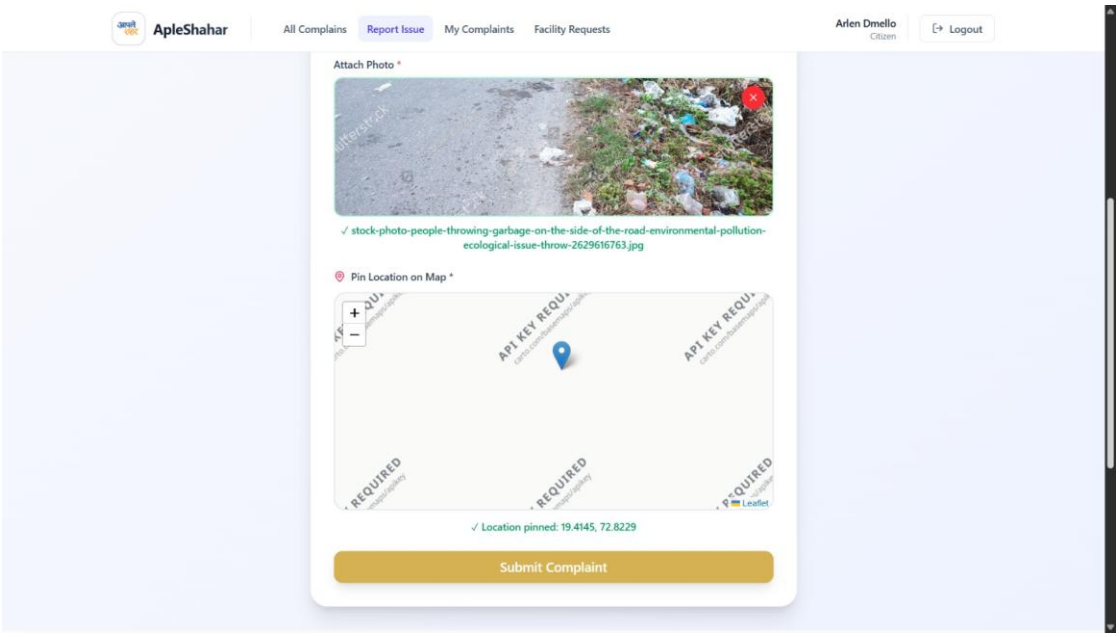


Figure 4.3: Interactive Leaflet Geolocation Pinning on Municipal Map

4.3 Public Grievance Feed & Active Work Zones

ApleShahar fosters public transparency through an open citizen feed displaying all reported civic issues. Interactive summary counters display Total, Pending, In Progress, and Resolved counts. Each complaint card displays the category, description, ward number, date reported, ticket ID, and evidence photograph.

The 'Active Work Zones' view plots complaints across Vasai-Virar on a dynamic GIS map with color-coded markers (orange for Pending, blue for In Progress, green for Resolved), providing citizens and ward supervisors with instant spatial awareness of civic trouble spots.

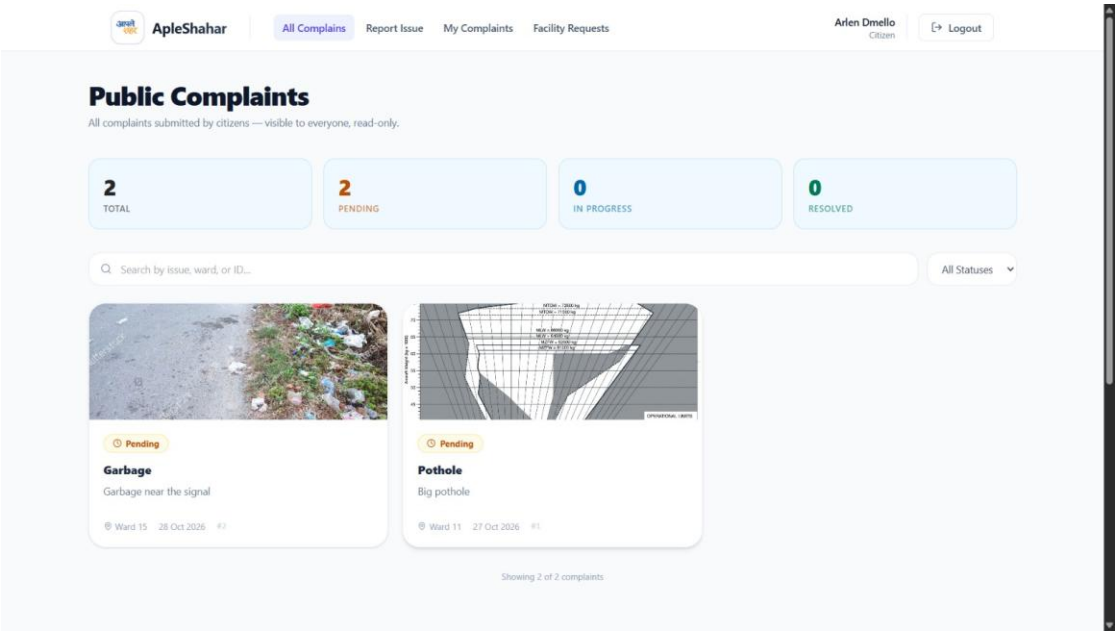


Figure 4.4: Public Grievance Feed with Real-Time Metric Counters & Cards

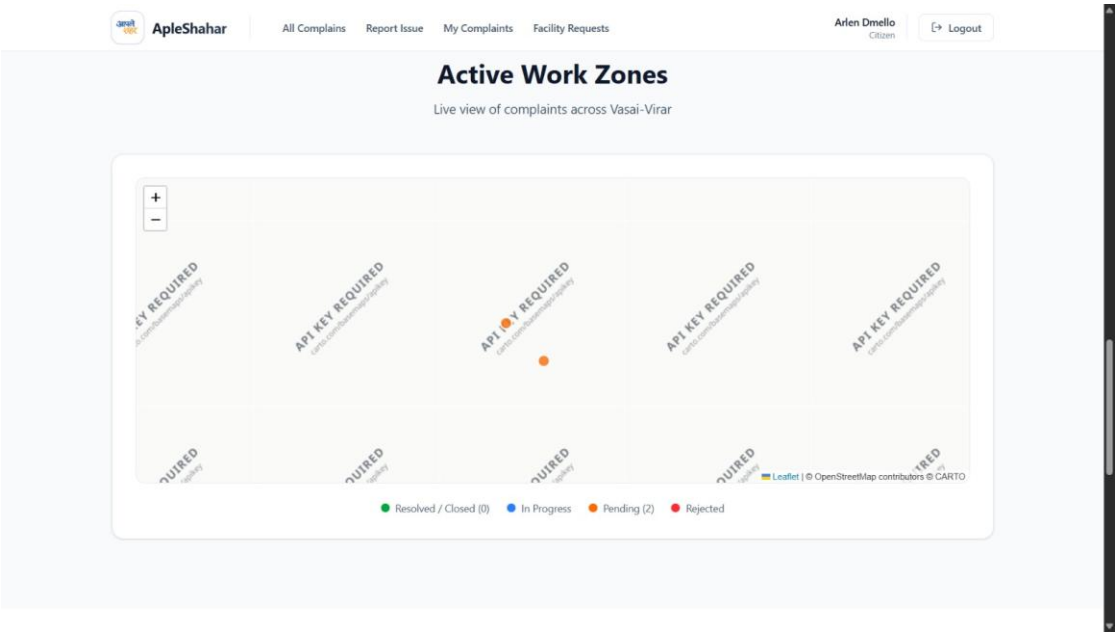


Figure 4.5: Active Work Zones Map with Color-Coded Status Markers

4.4 Citizen Grievance Lifecycle Tracking & Action Center

Citizens monitor their complaints via a detailed tracking view displaying the report ID, ward, issue category, GPS location, timestamp, and current lifecycle state. The Action Center informs the citizen of administrative progress and provides action buttons to verify resolution or request re-investigation once authorities mark the repair completed.

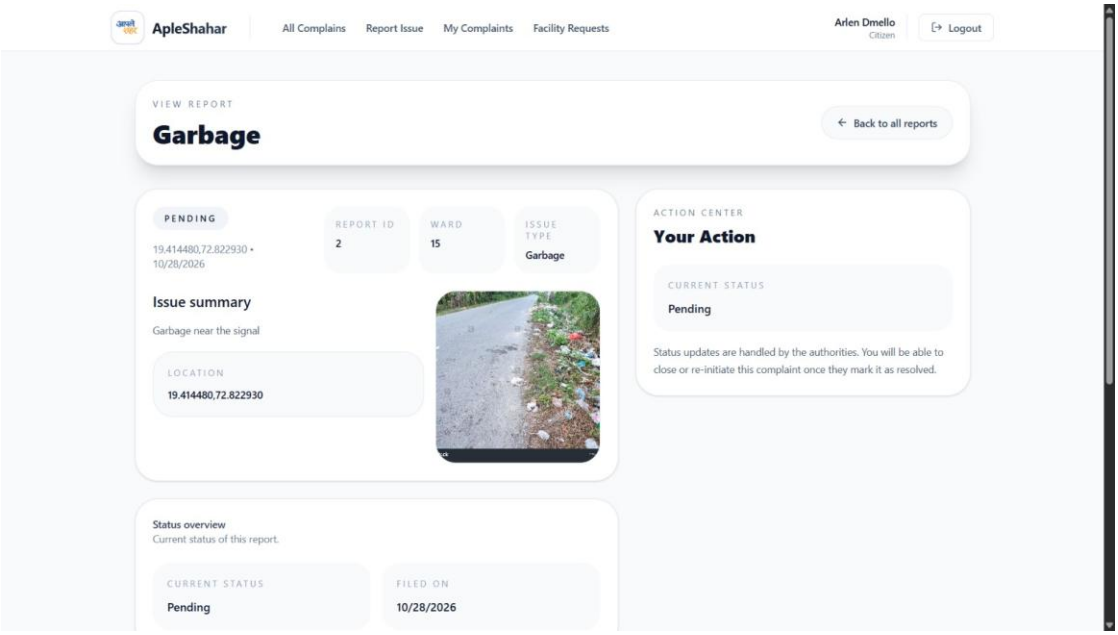


Figure 4.6: Citizen Grievance Detailed View & Progress Lifecycle Status

4.5 Community Amenity & Municipal Facility Proposals

To democratize urban infrastructure planning, citizens can submit amenity proposals (e.g., Public Benches, Waste Disposal Bins, Restrooms) needed in their neighborhood. Citizens provide justification, ward assignment, GPS coordinates, and site photos. Municipal authorities review community proposals in a dedicated modal, inspecting justification and photos before approving or rejecting proposals with official feedback.

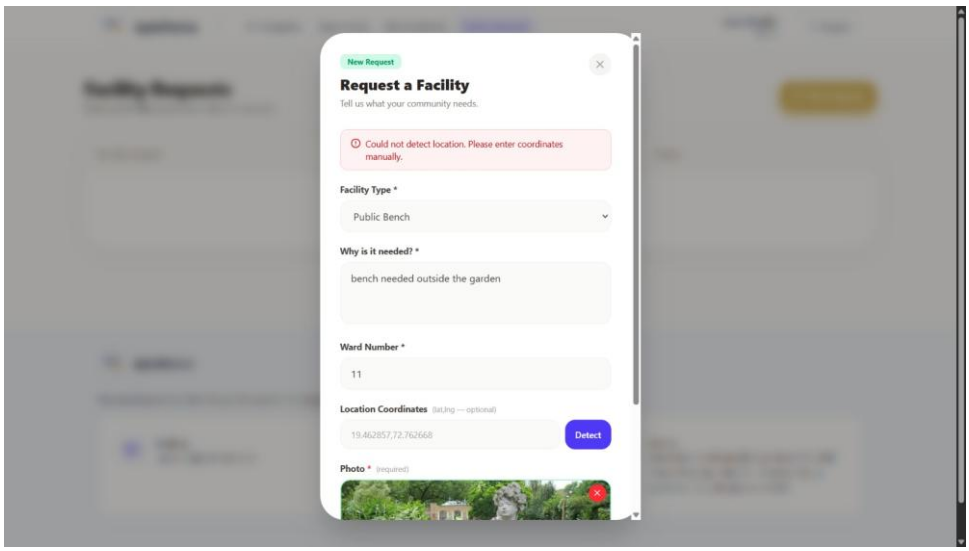


Figure 4.7: Citizen Community Amenity / Public Facility Proposal Modal

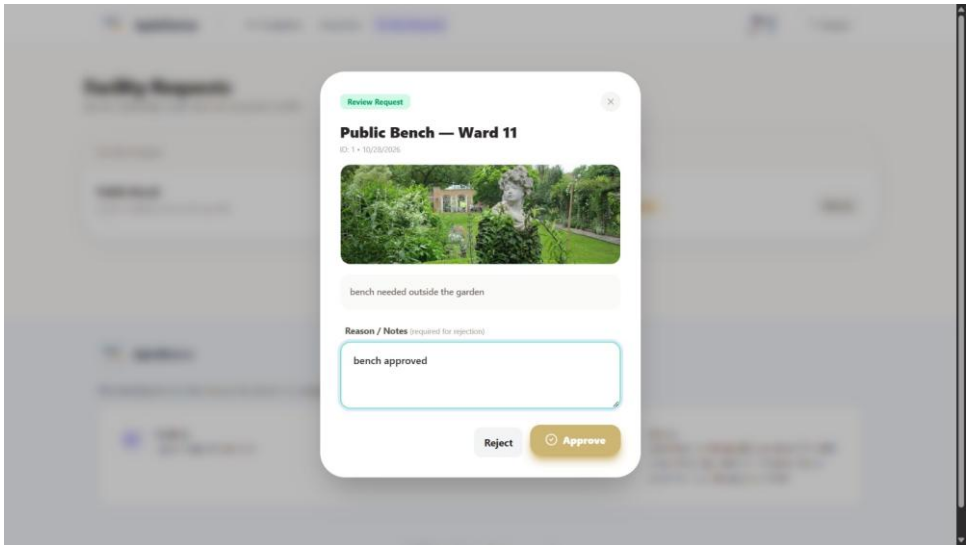


Figure 4.8: Municipal Authority Review & Decision Modal for Civic Amenities

4.6 Municipal Authority Management Portal & Repair Resolution

Municipal ward officers access an elevated management portal (e.g., Hansy, Authority). In the Action Center, authorities review citizen evidence, commit estimated resolution timelines (ETAs), update status to 'In Progress', and dispatch engineering crews.

Crucially, marking a complaint 'Resolved' requires entering a resolution description and attaching photographic proof of completed repairs (e.g., repaired road asphalt). Saving changes commits the resolution proof to S3, updates the PostgreSQL record, and triggers an automated Amazon SNS alert notifying the citizen to verify the physical repair.

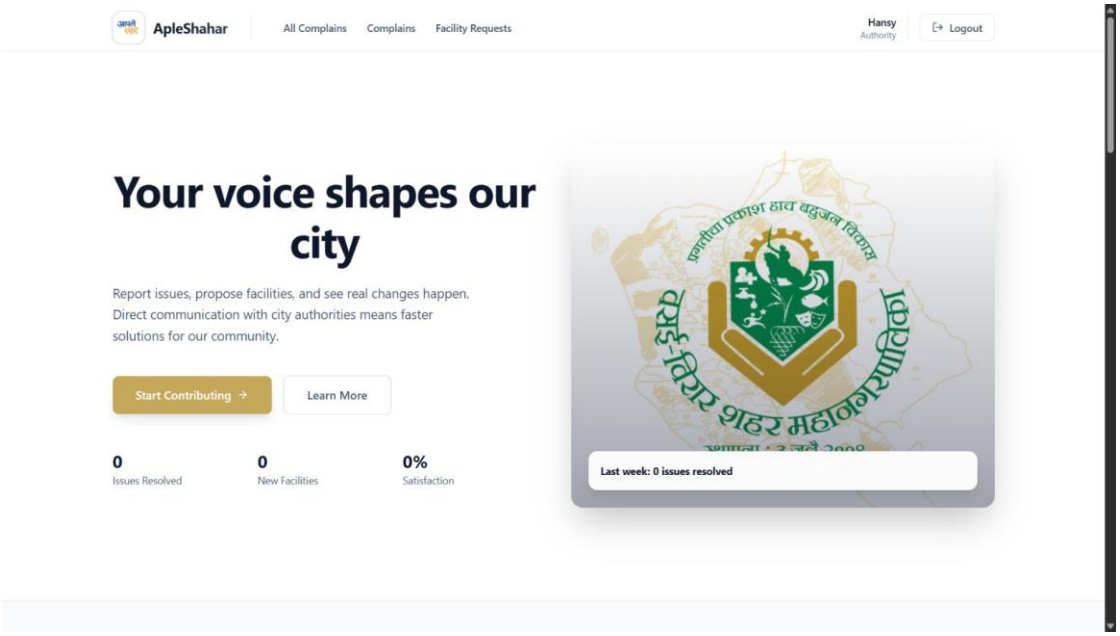


Figure 4.9: Municipal Authority Administrative Portal Interface

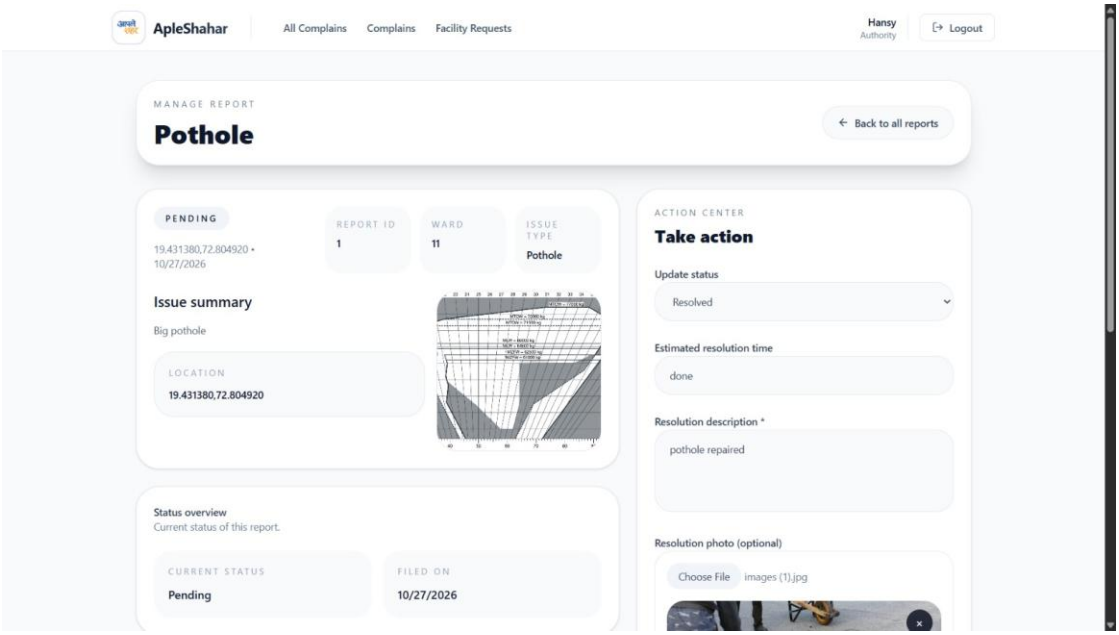


Figure 4.10: Authority Grievance Triage, Public ETA Commitment & Action Center

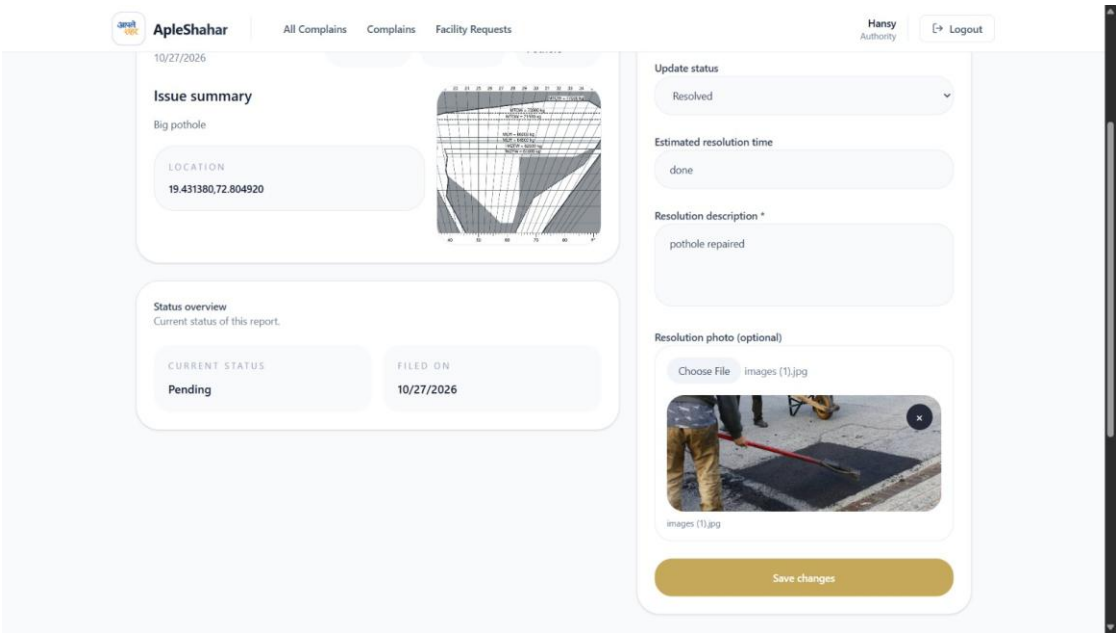


Figure 4.11: Completion Proof Photographic Evidence Upload & Ticket Closure

CHAPTER 5: OPERATIONAL COST BREAKDOWN & FREE TIER GOVERNANCE

5.1 Detailed Service-by-Service Cost Analysis

A primary engineering constraint of ApleShahar was designing a production-grade municipal architecture operating at $0 monthly cost under the AWS Free Tier. Table 5.1 provides a granular line-item cost breakdown of all provisioned cloud services.

Table 5.1: Monthly Cloud Infrastructure Cost & Free Tier Breakdown

AWS Service	Provisioned Resource Spec	Free Tier Allowance	Cost ($)
Amazon VPC	Custom VPC, 3 Subnets, IGW, Route Tables	Always Free (No NAT Gateway)	$0.00
Amazon EC2	1 x t3.micro (2 vCPU, 1 GB RAM, Ubuntu)	750 hours/month Free Tier	$0.00
Amazon RDS	PostgreSQL 16.10, db.t3.micro, 20GB gp3	750 hours/month + 20GB Storage	$0.00
Amazon S3 (Media)	Media bucket (complaints/, resolved/)	5 GB Standard Storage, 20K GETs	$0.00
Amazon S3 (Web)	Frontend static hosting bucket	5 GB Standard Storage, 2K PUTs	$0.00
Amazon CloudFront	Global CDN Distribution (E173DOOWG8BGP2)	1 TB Data Transfer Out/month	$0.00
AWS Lambda	ApleShahar-ImageResizer (512MB, Node 20)	1,000,000 requests/month Free	$0.00
Amazon SNS	Topic apleshahar-complaint-alerts	100,000 Free Email deliveries	$0.00
AWS SSM	Parameter Store (11 Standard parameters)	Standard parameters 100% Free	$0.00
Amazon CloudWatch	1 Metric Alarm + Lambda Log Groups	10 Alarms + 5 GB Logs Free	$0.00
AWS IAM & Budgets	Roles, Policies, $0.80 Budget Alert	Always Free Tier Included	$0.00

Total Estimated Monthly Operational Spend: $0.00 USD (100% Free Tier Compliant)

5.2 Architectural Free Tier Safeguards

Three deliberate architectural decisions prevent accidental cloud billing:

1. Elimination of NAT Gateways: Standard enterprise VPC designs deploy AWS NAT Gateways to grant private subnets internet access, costing ~$32/month. ApleShahar avoids this by hosting the containerized backend in an egress-enabled public subnet, communicating with RDS locally within the VPC while routing to S3, SNS, and SSM directly via the Internet Gateway at $0 cost.

2. Single-AZ RDS Deployment: Multi-AZ RDS replication is an enterprise paid feature. ApleShahar utilizes a Single-AZ deployment (encompassed by a multi-AZ subnet group for compliance) with automated daily snapshots, remaining 100% free.

3. Offloading CPU Resizing to Lambda: Sharp image processing on EC2 would force upgrading to a paid compute tier (e.g., t3.small or t3.medium). Offloading to serverless Lambda leverages the 1,000,000 free request monthly allowance.

5.3 Budget Threshold Governance ($0.80/Month Target)

To provide foolproof billing protection against runaway cloud consumption, an AWS Budget (ApleShahar-ZeroSpend-Budget) was created with a hard monthly threshold of $0.80 USD. If actual or forecasted monthly spend breaches 100% ($0.80), AWS Billing triggers an automated notification to apatade76@gmail.com, enabling immediate intervention before any credit depletion occurs.

CHAPTER 6: CONCLUSION & FUTURE SCOPE

6.1 Project Summary & Compliance Matrix

AppleShahar successfully establishes an enterprise-grade civic grievance redressal and municipal facility management platform engineered natively on Amazon Web Services. Table 6.1 verifies 100% compliance against all academic and cloud computing requirements.

Table 6.1: Academic & Cloud Computing Rubric Compliance Matrix

Cloud Computing Requirement	AppleShahar Implementation Details	Status
Mandatory Core Non-Serverless Compute	Node.js 20/Express Docker container on Amazon EC2 t3.micro	COMPLIANT
Mandatory Custom Virtual Private Cloud	Isolated VPC 10.0.0.0/16 with 1 public & 2 private subnets	COMPLIANT
Mandatory Managed Relational Database	Amazon RDS PostgreSQL 16.10 isolated in private subnets	COMPLIANT
Mandatory Cloud Object Storage	Amazon S3 for citizen grievance media and static web hosting	COMPLIANT
Zero Credential Hardcoding & Least Privilege	IAM Instance Profiles (IMDSv2) & SSM Parameter Store	COMPLIANT
Automated Cloud Monitoring & Alarms	Amazon CloudWatch alarm (CPU > 75%) routed to Amazon SNS	COMPLIANT
Financial Governance & Cost Control	AWS Budgets capped at $0.80/month with email threshold alerts	COMPLIANT
Supporting Capability 1: Serverless Compute	AWS Lambda ImageResizer (Sharp Node 20.x, ~82% compression)	COMPLIANT
Supporting Capability 2: Decoupled Messaging	Amazon SNS transactional pub/sub alerting for ward officers	COMPLIANT
Supporting Capability 3: Content Delivery	Amazon CloudFront multi-origin CDN with edge SSL termination	COMPLIANT

6.2 Key Technical Achievements

1. Complete Infrastructure as Code (IaC): Engineered end-to-end PowerShell (deploy-infra.ps1) and Bash (deploy-infra.sh) automation scripts provisioning networking, databases, storage, and IAM roles in a single execution.

2. Sub-20ms Media Latency: Leveraged CloudFront edge caching at Mumbai PoP (BOM78) and asynchronous Lambda Sharp image downsampling to reduce citizen image sizes from 608 KB to 107 KB.

3. Zero-Touch Database Initialization: Built automated DDL schema provisioning routines (initDb()) inside PostgreSQL connection pooling, guaranteeing instant schema setup on fresh database instances.

4. Transparent Civic Accountability: Enforced mandatory photographic verification and public ETA commitments before municipal tickets can be marked resolved.

6.3 Future Roadmap & Scalability

• Automated AI Defect Classification: Integrate Amazon Rekognition to automatically inspect uploaded citizen photos, verify genuine road potholes or garbage dumps, and reject irrelevant images at upload time.

• WhatsApp Municipal Chatbot: Connect Amazon Lex and AWS Lambda to the official VVCMC WhatsApp Business API, allowing citizens to lodge complaints directly via WhatsApp messaging.

• Predictive Municipal GIS Heatmaps: Implement spatial clustering algorithms to identify recurrent water pipeline bursts and road degradation zones, enabling predictive infrastructure budgeting.

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